AI Usage Disclosure

I used AI to understand the assignment TODOs. Especially TODOs of parts 6 and 7 of the assignment as they were a bit challenging for me as we had to work on our own data and get results from it. I used gemini.

Assignment 4 Report Answers

Part 1: Objectron Written Questions

1. The Two Intermediate Representations: The Objectron encoder gives us two main things: a heatmap and displacement fields.

   The heatmap is like a probability map that shows where the center of the object (like a chair) is most likely to be.

   The displacement fields provide the x and y offsets for the 8 corners of the 3D bounding box relative to that center.

   They work together because the heatmap finds the "anchor" (the center), and the displacements tell us the "shape" of the box.
2. How decode() Works: The decode() function first finds the peak (highest value) in the heatmap to locate the object's center. Then, it looks at that same spot in the displacement fields to get the 16 offset values for the 8 corners. By adding these offsets to the center point, the code calculates the exact 2D positions of all the bounding box vertices.
3. Fixed Resolution: The model is specifically trained for 480×640 images, so it won't work correctly with other sizes. To fix this, the code resizes any input image to 480×640 before feeding it to the model. It then gets normalized coordinates (0 to 1) and scales them back up to the original image's size so the result looks correct to the user.

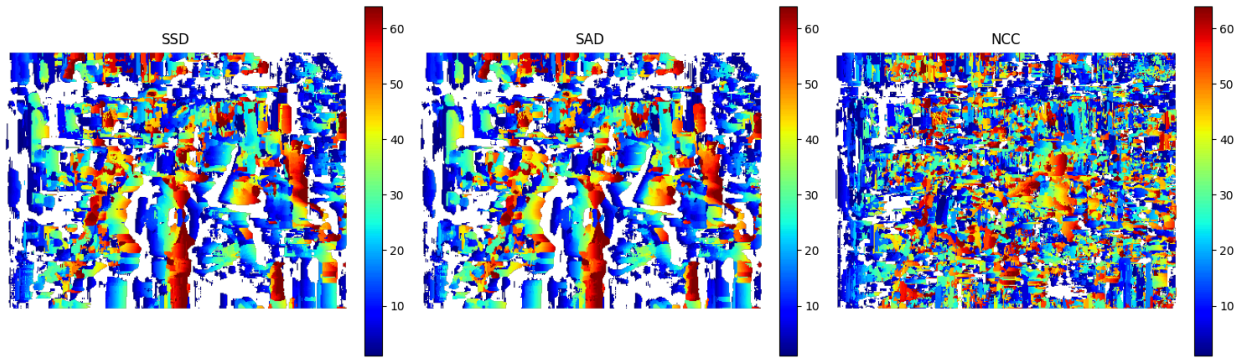
Part 3: Pose Estimation

1. Landmarks: MediaPipe detects 33 landmarks. These include the nose (index 0), left shoulder (index 11), and left wrist (index 15). Having this many points is great for AR or action recognition because it allows for very precise tracking of body movements, like rotation or specific limb poses, which wouldn't be possible with just a few points.
2. Normalized Coordinates: The model returns values between 0 and 1 so that it doesn't matter what resolution your image is. If it used pixels, you’d have to retrain the model for every different camera size. With normalized values, you just multiply them by the width and height of your specific image to get the pixel locations.
3. Thumb Tip vs. Wrist: Using the thumb tip (index 22) is better than the wrist (index 16) for detecting touches because the thumb is at the very front of the hand. When you reach for a chair,

your fingers enter the bounding box first. If we used the wrist, the detection would be late and look like your hand is already deep inside the object before it triggers.

Part 8: Stereo Matching Analysis & Questions

8.6 Parameter Analysis



Window Size: Small windows (like size 3) are very noisy and grainy because they don't have enough context. Large windows (like 21) look smooth but blur the edges between objects. Size 7 is the best "sweet spot" for noise and sharpness.

Disparity Range: A small max_disparity is fast but "clips" close objects, making them look far away. A large range captures everything but takes much longer to compute.

SSD vs. SAD vs. NCC: SAD was the best choice, it’s both faster and slightly more accurate than SSD because it doesn't use complex squaring. NCC was the worst; it was 9x slower and struggled with flat, uniform areas.

Shortcomings: This algorithm fails on areas with no texture (like flat walls), edges (where it smears), and occlusions (parts hidden from one camera)

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SSD avg time: 38.983s
SAD avg time: 34.429s
NCC avg time: 312.865s
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Running Time: Speed depends on image size, disparity range, and window size. NCC is far too slow for real-time use because of the extra math needed for every pixel.

8.1 - 8.3 Written Questions

1. Fundamental Matrix: When cameras aren't perfectly aligned, you can't just search horizontally for matches. The Fundamental Matrix (F) describes the relationship between the two views. It tells us that a point in the left image must lie on a specific line (the epipolar line) in the right image, which narrows the search from the whole image down to a single line.
2. Stereo Rectification: This process warps the images so the epipolar lines become perfectly horizontal. This is necessary because it simplifies the matching search to a single row, making it

much faster. It also makes the horizontal shift (disparity) easy to convert into actual depth values.

3. Depth with Unknown Pose: If the pose is unknown, you follow these steps:
    - Match features (like SIFT) between images.
    - Estimate the Fundamental Matrix (F) from those points.
    - Use camera calibration (K) to get the Essential Matrix (E).
    - Decompose E to find the rotation and translation (pose).
    - Rectify the images, run the window matching, and calculate depth from disparity.